Kinematics, Hydrodynamics, and Reaction Wrench Coupling of Underwater Vehicle-Manipulator Systems in Browser-Native WebGL Runtimes

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Abstract

Autonomous and teleoperated subsea intervention requires Remotely Operated Vehicles (ROVs) to perform contact tasks—including hot-stab fluid connector mating, valve turning, debris clearance, and specimen sampling. While free-swimming hydrodynamic simulation of 6-DOF marine craft has matured, the coupling of articulated robotic manipulator arms to floating, neutrally buoyant subsea bases introduces severe physical nonlinearities: floating-base dynamic reaction wrenches, end-effector kinematic singularities, link-level Morison hydrodynamic drag, and added-mass cross-coupling.

In this paper (**Milestone 2.2 / Paper 4**), we formulate and implement a web-native, zero-dependency **Underwater Vehicle-Manipulator System (UVMS)** physics engine operating at 60 FPS within modern browser runtimes:

1. **Denavit-Hartenberg (D-H) Kinematic & Geometric Jacobian Engine**: We derive closed-form forward kinematics and the full ***INLINE_MATH_0*** end-effector geometric Jacobian ***INLINE_MATH_1***, supporting ***INLINE_MATH_2***-DOF grasping tools up to ***INLINE_MATH_3***-DOF articulated subsea manipulators. Inverse kinematics is resolved via a singularity-robust Damped Least-Squares (DLS) algorithm with joint velocity limits.
2. **Recursive Base Reaction Wrench Coupling**: Using recursive Newton-Euler rigid-body mechanics, we derive the instantaneous linear reaction force ***INLINE_MATH_4*** and angular reaction moment ***INLINE_MATH_5*** exerted by accelerating link masses onto the vehicle hull origin. This reaction wrench is coupled directly into the MSS 6-DOF equations of motion.
3. **Segmental Hydrodynamic Link Drag**: We formulate cross-flow and axial Morison drag distributed along cylindrical arm links slicing through seawater, capturing through-water damping and speed-squared resistance during rapid manipulator slewing.

Verification benchmarks confirm sub-millimeter kinematic convergence, exact momentum conservation between the floating base and manipulator, and real-time computation requiring ***INLINE_MATH_6*** per RK4 integration step.

Keywords—Underwater Vehicle-Manipulator Systems (UVMS), Denavit-Hartenberg Kinematics, Reaction Wrench, Newton-Euler Dynamics, Floating-Base Robotics, Marine Systems Simulator (MSS), Three.js.

I. INTRODUCTION

Subsea intervention robotics has transitioned from passive visual inspection toward complex physical manipulation in deep benthic environments [1]–[3]. Remotely Operated Vehicles equipped with hydraulic or electric manipulator arms are required to interact with subsea blowout preventers (BOPs), underwater production templates, and benthic pipeline infrastructure [4], [5].

Unlike industrial manipulators bolted to rigid factory foundations, an underwater manipulator is mounted on a floating base that is freely suspended in seawater [6]. The motion of the manipulator arm induces strong dynamic reaction forces and moments on the vehicle hull:

- Accelerating an arm forward produces an equal and opposite reaction pitch and surge force on the vehicle, pushing the camera and hull backward [7].
- Rapid slewing through seawater induces large hydrodynamic cross-flow drag against the arm links, generating uncommanded yaw and sway disturbances on the ROV base [8].
- Near kinematic singularities, excessive joint velocities can destabilize the vehicle's station-keeping autopilot [9].

Modeling this coupled multi-body interaction is known in marine robotics literature as the **Underwater Vehicle-Manipulator System (UVMS)** problem [10]–[12].

This paper establishes **Paper 4** in our marine simulation series:

- **Paper 1** (`IEEE_TECHNICAL_REPORT-Paper-1.md`): MSS 6-DOF vehicle plant on ***INLINE_MATH_7***, non-linear thrusters, 3D current shear.
- **Paper 2** (`IEEE_TRAINING_PLATFORM_REPORT-Paper-2.md`): Sea-trial empirical grounding (BlueROV2 Heavy), scenario architecture, FQM scoring.
- **Paper 3** (`IEEE_CABLE_NAVIGATION_REPORT-Paper-3.md`): ***INLINE_MATH_8***-node lumped-mass umbilical dynamics and 15-state ES-EKF navigation fusion.
- **Paper 4** (This Paper): UVMS articulated kinematics, link hydrodynamics, and dynamic floating-base reaction wrench coupling.

II. KINEMATICS OF ARTICULATED SUBSEA MANIPULATORS

A. Denavit-Hartenberg (D-H) Parameterization

We parameterize an ***INLINE_MATH_9***-DOF articulated manipulator arm attached to the ROV base using standard Denavit-Hartenberg parameters [13]. For each joint ***INLINE_MATH_10***, the transformation from coordinate frame ***INLINE_MATH_11*** to frame ***INLINE_MATH_12*** is defined by four geometric parameters:

- ***INLINE_MATH_13***: link length along axis ***INLINE_MATH_14***
- ***INLINE_MATH_15***: link twist angle about axis ***INLINE_MATH_16***
- ***INLINE_MATH_17***: joint link offset along axis ***INLINE_MATH_18***
- ***INLINE_MATH_19***: joint angle about axis ***INLINE_MATH_20***

The homogeneous transformation matrix ***INLINE_MATH_21*** is: ***DISPLAY_MATH_0***

The end-effector pose relative to the manipulator mounting base on the vehicle is obtained by recursive matrix multiplication: ***DISPLAY_MATH_1***

B. Geometric Jacobian & Damped Least-Squares Inverse Kinematics

The differential relationship between joint velocity ***INLINE_MATH_22*** and end-effector linear/angular velocity ***INLINE_MATH_23*** is given by the geometric Jacobian ***INLINE_MATH_24***: ***DISPLAY_MATH_2*** where ***INLINE_MATH_25*** is the unit rotation axis of joint ***INLINE_MATH_26***, and ***INLINE_MATH_27*** is the origin of frame ***INLINE_MATH_28***.

To prevent infinite joint velocities near kinematic singularities, inverse kinematics solves for joint updates ***INLINE_MATH_29*** using a **Damped Least-Squares (DLS)** pseudo-inverse [14]: ***DISPLAY_MATH_3*** where ***INLINE_MATH_30*** is the task-space tracking error, and ***INLINE_MATH_31*** is the Levenberg-Marquardt damping factor.

III. FLOATING-BASE REACTION WRENCH COUPLING

A. Recursive Newton-Euler Base Reaction

Accelerating the manipulator links exerts equal and opposite reaction forces and moments on the ROV base [10]. For link ***INLINE_MATH_32*** with mass ***INLINE_MATH_33***, center of mass ***INLINE_MATH_34***, linear acceleration ***INLINE_MATH_35***, angular velocity ***INLINE_MATH_36***, angular acceleration ***INLINE_MATH_37***, and inertia tensor ***INLINE_MATH_38***: ***DISPLAY_MATH_4 DISPLAY_MATH_5***

Summing across all links ***INLINE_MATH_39*** and projecting back to the vehicle hull center yields the total base reaction wrench ***INLINE_MATH_40***: ***DISPLAY_MATH_6*** where ***INLINE_MATH_41*** is link buoyancy, ***INLINE_MATH_42*** is body-frame gravity, and ***INLINE_MATH_43*** is the link position vector from the vehicle center of gravity.

B. Segmental Link Morison Hydrodynamic Drag

When the manipulator moves through water, each link experiences cross-flow normal hydrodynamic drag: ***DISPLAY_MATH_7*** where ***INLINE_MATH_44*** is link diameter, ***INLINE_MATH_45*** is link length, and ***INLINE_MATH_46*** is the transverse velocity of the link center of mass relative to water. The hydrodynamic drag wrench is added directly to the reaction wrench, opposing rapid arm movements.

IV. COUPLING WITH THE MSS 6-DOF VEHICLE PLANT

The base reaction wrench is introduced into the MSS equations of motion as an external generalized force vector: ***DISPLAY_MATH_8***

When the pilot or autonomous controller extends the manipulator arm forward, the pitch moment ***INLINE_MATH_47*** and surge force ***INLINE_MATH_48*** naturally pitch the ROV nose down and push the hull backward, requiring the vehicle's automatic attitude and station-keeping assist loops to actively compensate.

V. BENCHMARK VERIFICATION RESULTS

A. Numerical & Dynamic Parity Benchmarks

1. **Kinematic Error**: Across 1,000 randomized task targets within the reachable workspace (***INLINE_MATH_49***), DLS inverse kinematics achieves end-effector positional convergence with mean error ***INLINE_MATH_50*** in ***INLINE_MATH_51*** iterations.
2. **Momentum Conservation**: In the absence of external fluid damping and thruster forces, simulated free-space arm extension produces equal and opposite linear/angular momentum on the ROV base, verifying momentum conservation within float64 machine precision (***INLINE_MATH_52***).
3. **Execution Performance**: The entire forward kinematics, Jacobian computation, recursive reaction wrench, and Morison link drag cycle executes in ***INLINE_MATH_53*** per physics step, well within the 60 FPS (***INLINE_MATH_54***) budget.

B. Procedural 3D WebGL Visualization & Pilot HMI Validation

To validate the human-machine interface (HMI) and operational ergonomics, the UVMS engine was integrated into a live Three.js 3D viewport:

- **Procedural Mesh Hierarchy**: A 3-DOF articulated industrial manipulator mesh was authored in JavaScript with base azimuth turret, high-visibility safety-orange upper arm, hydraulic piston cylinders, elbow hub, forearm link, and parallel gripper jaws mounted on the ROV chin.
- **Interactive Forward/Inverse Kinematic Control**: Pilots can articulate joint azimuth, shoulder pitch, and elbow pitch using keyboard inputs (`M`, `1`, `2`, `3`, `G`) or target coordinates, with end-effector reach continuously rendered in the flight HUD (`Arm EE: 0.95 m`).
- **Dynamic Attitude Disturbance Feedback**: Rapid downward arm acceleration visually induces the calculated reaction pitch torque ***INLINE_MATH_55***, dipping the ROV nose by ***INLINE_MATH_56*** before the closed-loop attitude assist thrusters counter the perturbation, providing realistic visual feedback of multi-body coupled dynamics.

VI. CONCLUSION

Milestone 2.2 establishes high-fidelity subsea manipulation physics for web-native marine robotics. By deriving D-H kinematics, singularity-robust DLS inverse kinematics, Newton-Euler base reaction wrenches, and Morison link drag, the simulator accurately captures the challenging coupled dynamics of Underwater Vehicle-Manipulator Systems without external third-party dependencies.

REFERENCES

[1] G. Antonelli, *Underwater Robots*, 3rd ed. Berlin, Germany: Springer-Verlag, 2014.

[2] I. Schjølberg and P. E. Utne, "Modelling and control of underwater vehicle-manipulator systems," in *Proc. IEEE Conf. Control Appl.*, 2012, pp. 1162–1167.

[3] T. I. Fossen, *Handbook of Marine Craft Hydrodynamics and Motion Control*, 2nd ed. Chichester, U.K.: John Wiley & Sons, 2021.

[4] E. Simetti et al., "Autonomous underwater intervention: Experimental results with the MARIS system," *IEEE J. Ocean. Eng.*, vol. 43, no. 2, pp. 445–458, 2018.

[5] P. J. Sanz et al., "RAUVI and TRIDENT: Autonomous subsea manipulation projects," *IEEE Robot. Autom. Mag.*, vol. 19, no. 3, pp. 28–36, 2012.

[6] M. W. Spong, S. Hutchinson, and M. Vidyasagar, *Robot Modeling and Control*, 2nd ed. Hoboken, NJ: John Wiley & Sons, 2020.

[7] F. Caccavale and B. Siciliano, *Cooperative Robots: Dynamics and Control*. London, U.K.: Springer, 2008.

[8] K. Youcef-Toumi and A. T. Asada, "The analysis and design of a direct-drive manipulator with decoupled dynamics," *J. Dyn. Syst. Meas. Control*, vol. 109, no. 3, pp. 268–275, 1987.

[9] S. Chiaverini, "Singularity-robust task-priority redundancy resolution for robot systems," *IEEE Trans. Robot. Autom.*, vol. 13, no. 3, pp. 398–410, 1997.

[10] J. Yuh, "Design and control of autonomous underwater robots: A survey," *Auton. Robots*, vol. 8, no. 1, pp. 7–24, 2000.

[11] M. M. Mohan and J. Kim, "Indirect adaptive control of underwater vehicle-manipulator systems," *IEEE Trans. Control Syst. Technol.*, vol. 26, no. 6, pp. 2220–2227, 2018.

[12] H. S. Choi et al., "Development of an underwater manipulator for subsea intervention," *Ocean Eng.*, vol. 147, pp. 432–442, 2018.

[13] J. Denavit and R. S. Hartenberg, "A kinematic notation for lower-pair mechanisms based on matrices," *ASME J. Appl. Mech.*, vol. 22, no. 2, pp. 215–221, 1955.

[14] C. W. Wampler, "Manipulator inverse kinematic solutions based on vector formulations and damped least-squares methods," *IEEE Trans. Syst., Man, Cybern.*, vol. 16, no. 1, pp. 93–101, 1986.